

Effects of Postactivation Potentiation Warm-up in Male and Female Sport Performances: A Brief Review

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SUMMARY

A PROPER PRECOMPETITIVE WARM-UP DURING SPRINT COMPETITION WITH EXPLOSIVE ATHLETES WILL ELICIT A POSTACTIVATION POTENTIATION (PAP). THIS ARTICLE WILL REVIEW THE EFFECTS OF HEAVY PRELOAD WARM-UP PROTOCOLS ELICITING PAP WITH EXPLOSIVE MALE AND FEMALE ATHLETES, RESULTING IN ENHANCED PERFORMANCES AND PROVIDE STRENGTH COACHES WITH PRACTICAL APPLICATIONS.

INTRODUCTION

Collegiate and professional sport coaches and strength coaches warm-up their respective male and female competitive athletes before all practices and competitive events. All these athletes and strength coaches seek the proper warm-up regimens to hopefully improve optimum competitive performances. Based on proven warm-up research, strength coaches need to educate explosive athletes that if they perform the proper specific warm-up protocols, they may increase their performances and reduce the high risk of injury during their ballistic events. Past and current research

studies have reported that sport-specific resistance warm-up increases performances in explosive activities such as baseball hitting and pitching (7,8,48).

Likewise, recent investigators have reported explosive jump performance increases after high-intensity (noncontrast or noncomplex training) preload resistance warm-up of short duration (2,7,9,12,15,25,35,37,42,50). This temporary muscle performance increase has been attributed to a phenomenon known as postactivation potentiation (PAP), a condition by which acute muscle force output is enhanced as a result of contractile history (36,40). If a precompetitive warm-up consists of the proper protocol and is conducted in a timely precise manner before the explosive event, then a PAP effect maybe elicited, and there is a possibility that jump and sport performances may increase. In addition, past PAP warm-up research indicates equivocal performance findings (1,4,5,9,10,13,16,20,22,23,25–27, 35,37,38,49,50).

In general, this brief review will focus on the past and present PAP warm-up research with elite male and female explosive athletes. Specifically, the purposes of this review are to investigate past and present PAP warm-up studies

to determine if PAP effects (a) are gender specific and (b) enhance explosive performances in male and female elite athletes.

INFLUENCES ON PAP

During explosive warm-up, the possibility of muscle fatigue can negatively affect contractile history and impairs force production or power output (6,17,32,36,39), thus decreasing explosive performance within a certain time. Therefore, some investigators have reported jump performance decreases (20,23,24,38,41) after preload high-intensity warm-up, which has been attributed to possible muscle fatigue. Nevertheless, research investigators have reported a coexistence of fatigue and PAP in skeletal muscle (21,36) and that muscle performance enhancement after preload warm-up depends on the balance between muscle fatigue and muscle potentiation (14,29,40,48). Optimal performance occurs when fatigue has subsided but PAP effect still exists (21).

In addition to fatigue, it has been reported that there are other possible influences on PAP: muscle fiber type

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(1,16,19,39), performance level (15,51), training regimens (9), type of exercise (35,42), recovery time between the preload stimulus and the dynamic performance testing (25), training experience (5,16,37), gender (13,17,37), and intensities (6). Although the mechanisms for eliciting PAP and subsequent improved performances are uncertain, the 2 most reasonable underlying PAP mechanism theories include increased phosphorylation of light chain myosin and increased neurological factors in the spinal cord (38,40). Hodgson et al. (21) summarized PAP studies, which attribute primarily the PAP phenomenon to the physiological events within the muscle, such as the phosphorylation of myosin light chains (1). This small protein improves the interaction between myosin and thin filaments. This phosphate binding leads to an increase in the rate by which myosin cross-bridges move from nonforce to force producing resulting in contractile muscle activation. In addition, PAP possibly occurs in the spinal cord through an increased synaptic efficacy between Ia afferent and α -motoneurons of the homonymous muscle. Moreover, possibly, PAP occurs as a result of both myogenic and neurogenic influences (21).

MALE PAP STUDIES

The majority of male PAP warm-up studies have been conducted in lower-body jump performances (horizontal and vertical) (4,9,11,13,16,23,25–27, 35,38,41,48) and upper-body performances in bench press throws, medicine ball power drop, and explosive push-ups (1,10,16,22). These warm-up studies reported increased jump and upper-body performances after a PAP protocol. In contrast, the following studies reported a decrease in explosive push-ups (22) and jump performances (9,23,26,27,38,48) after a PAP warm-up protocol. Thus, past PAP research has indicated equivocal male performance findings.

Contrary to male non-sport-specific jump and upper-body PAP studies, male sport-specific PAP warm-up studies reported increases in dynamic

performances in competitive sports such as cycling and sprinting (3,30,31,34,43,46). Chatzopoulos et al. (3) reported the influence of PAP on short-distance sprints of 10 and 30 m when performed by male elite amateurs in various sports after heavy 10 single repetition squats at 90% of 1 repetition maximum (RM). McBride et al. (31) also reported a PAP presence with collegiate football players on 40-m sprint performances after a low-volume, heavy squat warm-up consisting of 1 set of 3 repetitions at 90% of players' 1RM. Pfaff (34) found significant increase in performances with male elite sprinters performing a heavy resistance warm-up consisting of 90% of their 1RM for 5 sets of 1 repetition in the back squat exercise. In addition, elite rugby male players had significantly improved 20-m sprint times after a preload back squat warm-up consisting of 1 set of 1 repetition at 5RM (30). Moreover, Smith et al. (43) reported a significant improvement in male cycle sprint times using a modified Pfaff heavy preload warm-up consisting of 10 reps $\times$ 1 set at 90% 1RM. In contrast, recently, Till and Cooke (46) reported no significant group PAP effect on sprint (10 and 20 m) and counter-movement vertical jump performances after dynamic and isometric maximum voluntary contractions (MVC) with elite professional soccer players (Table).

FEMALE PAP STUDIES

To date, there have been limited studies examining PAP effects in female sport performances, and most of these studies have examined jump performances (5,9,16,20,37). In one of the first male and female PAP warm-up studies, Gullich and Schmidtbleicher (16) reported significant countermovement vertical jump height increases of 3.3% for both men and women after isometric voluntary squats warm-up. In a related lower-body warm-up to PAP, O'Leary et al. (33) reported no gender differences in twitch torque potentiation over the 5-minute posttetanus. The investigators suggested that in the dorsiflexor muscles, men and women show a similar amount and pattern of

twitch force potentiation. Therefore, based on the results of O'Leary et al. (33), the investigators in Chiu et al. (5) PAP study pooled their male and female subjects for analysis. This study (5) reported significantly greater increases in concentric-only jump squat performances in the athletic male and female groups ($n = 7$) than in the recreationally trained male and female groups ($n = 17$). The subjects in the elite athletic group were all involved in explosive strength sports: 1 Division I NCAA soccer player, 1 elite national triathlete, and 5 U.S.A. Weightlifting Collegiate National Champions. The investigators' results (5) indicated that athletic individuals respond more favorably to PAP than recreationally trained individuals.

In contrast, in another jump study, Hanson et al. (20) suggest that regardless of a warm-up of a single light-load, heavy-velocity squat intervention or a heavy-load, low-velocity squat intervention, there were no significant jump increases with either male or female collegiate resistance-trained subjects. The investigators suggest that the lack of jump improvements may be attributed to the PAP stimulated by the heavy-load low-velocity squat intervention set (SIS) being insufficient in magnitude or dissipated before post-testing. In addition, the lack of jump improvements also may be because of the submaximal 30% workload during SIS, insufficient movement pattern specificity between squat exercises and the countermovement vertical jump, or excessive rest intervals.

In another male-female PAP jump warm-up study with experienced and inexperienced weightlifters, Rixon et al. (37) investigated the type of muscle contraction (isometric versus dynamic), gender, and lifting experience on PAP as changes in jump height and power output. Thirty men ($n = 15$) and women ($n = 15$) classified as either experienced or nonexperienced weightlifters performed 3 different countermovement jumps, with the first set (pretest) used as baseline of jump height and power. The second set was

performed after a maximal isometric squat protocol (MVC-PAP), and the third set of jumps was performed after a maximal dynamic squat (DS) protocol (DS-PAP). Rixon et al. (37) reported that after the MVC-PAP protocol, jump height was significantly higher than both the pretest and DS-PAP. In addition, men performed significantly better than women, and experienced weightlifters performed significantly better than inexperienced weightlifters. Similarly, jump power was also significantly greater for MVC-PAP warm-up than for the other 2 conditions, and DS-PAP power improved as compared with pretest, with men performing significantly better than women. The investigators concluded that the isometric MVC-PAP treatment evoked a greater muscle PAP than the dynamic condition (DS-PAP), and postactivation was enhanced by previous weightlifting experience. The investigators concluded that the manipulation of MVC by pushing, squatting, or both against fixed objects, such as walls and low ceilings, could be a cost-effective and simple way to arouse a PAP state before sport performances.

Duthie et al. (9) compared jump performances after 3 different heavy-load and light-load training methods (PAP warm-ups) for 3 different training sessions using only female subjects. The women ($n = 11$) regularly participated in resistance training for their respective sport (hockey and softball). All women had been involved in high-intensity resistance training for more than 2 years. The half squat was used as the heavy loaded exercise, and jump squats were performed as the light-load exercise. Three sets of both the half squats and the jump squats were performed during each session. PAP 1 warm-up consisted of the traditional method of completing sets of power exercises (jump squats) before sets of half squats. PAP 2 warm-up consisted of completing sets of half squats before jump squats (complex method). PAP 3 involved alternating sets of half squats and jump squats (contrast method). In

contrast to the majority of male jump PAP warm-up studies, the main result in the study by Dutie et al. (9) revealed that there was no significant difference in jump squat performances (mean jump height, peak power, or maximal force) between each PAP warm-up method. In addition, there was a significant difference ($p < 0.05$) in performance between the higher and lower strength groups, with the higher strength group having a greater improvement in performance using the contrast PAP as compared with the traditional PAP. The investigators concluded that contrast training (PAP warm-up) is advantageous for increasing power output but only for athletes (women) with relatively high strength levels.

Recently, in a sport-specific PAP warm-up study, Linder et al. (28) investigated the effects of PAP on track sprint performance after a preload set of 4RM parallel back half squat exercises in collegiate women. All subjects ($n = 12$) participated in 2 testing sessions over a 3-week period. During the first testing session, subjects performed the *controlled* protocol consisting of a 4-minute standardized warm-up, followed by a 4-minute active rest, a 100-m track sprint, a second 4-minute active rest, finalized with a second 100-m sprint. The second testing session, the treatment protocol, consisted of a 4-minute standardized aerobic warm-up, followed by 4-minute active rest, sprint, a second 4-minute active rest, a preload heavy warm-up of 1 set of 4 repetitions with a 4RM load with parallel back half squats, a third 9-minute active rest, finalized with a second sprint. The results indicated that there was a significant improvement of 0.19 seconds ($p < 0.05$), when the second sprint was preceded by a 4RM half back squat protocol during *treatment*. Additionally, the results indicated that it would be expected that mean sprint times would increase by 0.04–0.34 seconds ($p < 0.05$), when using a preload 4RM squat protocol. The findings suggest that performing a 4RM parallel back half squat warm-up before a track sprint will have a positive PAP effect on decreased

track sprint times. The investigators suggest that strength and track coaches, looking for the “competitive edge” (PAP effect), may re-warm-up their sprinters during meets.

Linder et al. (28) support the male warm-up sport-specific studies (3,25, 30,31,34,43) that indicated a PAP effect after preload heavy resistance warm-up on short-distance sprints and sprint cycling performances. The study by Linder et al. (28) is unique because there are no other known female research studies on PAP effects after a preload warm-up on track mid distance sprints of over 40 yards. Future PAP warm-up research in competitive track sprints is needed (Table).

ELITE VERSUS RECREATIONAL ATHLETES

If a PAP effect is present, elite male or female athletes produce greater increases in jump or sport-specific performances than recreational athletes and physical education students (16). In the female studies of Duthie et al. (9) and Linder et al. (28), the elite and stronger female athletes performed greater on sport performances after a heavy preload PAP warm-up. Likewise, in the combination male and female PAP study of Chiu et al. (13), the athletic individuals responded more favorably to PAP than the recreationally trained individuals; and in the study of Rixon et al. (37), although the men performed significantly better than women, the experienced weightlifters (men and women) performed significantly better than the inexperienced weightlifters. As in the female PAP studies, in numerous male PAP studies (2,5,18,30,31,37,43), elite athletes increased sport-specific performances after a heavy preload PAP warm-up.

In addition, Smith et al. (43) reported that recreationally active men will not benefit from the effects of PAP during performance with a 7-minute recovery period. The investigators (43) support the previous PAP warm-up research that suggests that training status, strength, and skill level might be the keys in determining positive benefits of

PAP warm-up. In contrast, as previously stated, Hanson et al. (20) reported that there were no significant jump increases with either male or female collegiate resistance-trained subjects after a single light-load, heavy-velocity or a heavy-load, low-velocity squat PAP intervention. Yet, it appears that with the proper magnitude and/or increased submaximal workloads (5,9,28) or with adequate recovery (8–12 minutes) (25,28), the subjects of Hanson et al. (20) might have had improved jump performances.

SPRINT RESULTS

As was previously discussed in the male PAP studies section, there have been only a few heavy preloaded resistance warm-up PAP studies that have reported significant sprint improvements in swimming, track events, and cycling (3,28,30,31,34,43). Strength coaches should recognize that it appears in these sprint studies that a PAP effect after a preload heavy resistance stimulus may be responsible for the significant improvements (5,16,18,25,30,34,35,41,47,48), and this PAP effect may be more prominent for dynamic (concentric contractions) than isometric contractions (15,19). In contrast, Till and Cooke (46) conducted a PAP study with 12 elite male soccer players. The players performed 4 testing sessions over a 4-week period. All players were randomized into 4 groups, 3 experimental and 1 control. The 2 dynamic warm-up groups performed either the dead lift warm-up exercise or the double-legged tuck jump exercise, and the third group performed the isometric knee extensor exercise on the isokinetic dynamometer. There was no PAP treatment for the control group. After a controlled warm-up (5-minute jog) and the specific warm-up protocol, three 20-m sprints at 4, 5, and 6 minutes post-PAP were performed. In addition, after the sprint tests, the players performed 3 countermovement jumps at 7, 8, and 9 minutes post-PAP treatment. Again, the findings revealed no significant group PAP effect on sprint and jump performances.

MECHANICAL RESULTS

As strength coaches re-warm-up explosive athletes before the competitive sprinting events, mean power output will improve (43). If strength coaches re-warm-up using PAP, then maintaining high average power during sprinting events should transfer into faster times. Again, strength coaches must understand the effects of fatigue in PAP warm-ups. Previous PAP studies have reported that fatigue influences a decrease in force or power output (4,6,17,32). However, Smith et al. (43) reported that peak power and fatigue index did not significantly decrease with either a 5-minute or 20-minute rest recovery. Linder et al. (28) support the findings of Smith et al. (43) in that it is reasonable to assume that the 4RM parallel back half squat warm-up exercise did not elicit short-term anaerobic fatigue. The 2-minute rest period in between the squat exercises and the 9-minute active rest recovery post-4RM squat exercise appear to be adequate time for ATP resynthesis.

The PAP mechanisms for both men and women may be phosphorylation of myosin light chains during MVC (21,45). In addition, if muscle hypertrophy and fast fiber type conversions in heavy resistance-trained women are similar to men (44), then PAP effects within muscle may also be similar in women and men. It is still unclear as to the exact reason behind the relationship between strength and potentiation. However, it has been demonstrated that resistance-trained male and female athletes have greater activation of the musculature involved during a preload high-intensity stimulus, which would affect the H-reflex and myosin regulatory light chain phosphorylation, the 2 mechanisms involved in PAP (4).

RECOVERY

The results in Linder et al. (28) further indicate a PAP effect after the 9-minute recovery. Previous studies have determined the recovery period between the preload stimulus and the subsequent explosive performance activity that ranges from 0 to 18.5 minutes for

a phosphocreatine resynthesis after the preloaded warm-up (1,5,9,14,21,25,50). Kilduff et al. (25) reported that the optimal recovery to maximize the PAP effect on peak power output (PPO) (approximately 7–8% increases) needed to be between 8 and 12 minutes for the lower body. In contrast, Jones and Lees (24) reported no significant differences in any of the performance variables measured after 3, 10, and 20 minutes of recovery after preload squats. In addition, Gilbert and Lees (12) reported equivocal acute effects after preloading with no significant differences between pre- and post-weightlifting assessments of maximum isometric rate of force development at 2- and 10-minute recovery; in contrast, significant increases were reported at 15- and 20-minute recovery after preload repeated maximum strength (5 times at 1RM) back squats.

Although not unequivocal, the results of Linder et al. (28) suggest that the PAP warm-up may result in significant improvement in sprint times for 100-m sprint. Because the PAP warm-up did not result in all sprinters improving their times, there is the possibility that there are individual differences and particular conditions that need to be considered. Thus, PAP warm-up of Linder et al. (28) may not be appropriate for every sprinter.

CONCLUSIONS

Although PAP research reveals equivocal findings mainly in jump performances, as the recent PAP sport performance findings reviewed by the strength coaches indicate (a) a PAP effect was probably responsible for significant increases in swimming, cycling, and track sprint performances (3,28,30,31,34,43); (b) strong athletic-trained collegiate women revealed significant improvements in track sprints similar to trained men after a PAP intervention of heavy preload resistance warm-up; (c) to elicit a PAP effect for elite men, the heavy preload warm-up protocol should consist of 3–10 sets of 1 repetition at a load of 90% 1RM with isometric squats or dynamic parallel half back squats; (d) for elite

Table
Warm-up recommendations to elicit effects

Performers	Sports	Preload warm-up exercise	Rest interval (recovery)	Results
Men				
Jumpers	Vertical horizontal	Half squats: 3–5 sets, 1–3 repetitions of 60–90% of 1RM	3 min	Increases
		Back squats: 1–3 sets, 1–5 repetitions at 80–90% of 1RM	5–10 min	Increases
		Squat jumps countermovement jumps	3–5 min	Equivocal equivocal
Elite	Rugby	Back half squat: 1 set, 1–3 repetitions at 80–90% 5RM	5–10 min; 10 min	Increases 20-m sprint, 30-m sprint
		Back squat: 1 set, 5 repetitions, at 100% of 5RM	5 min	Increases 20-m sprint
		Bench press throws: 1 set, 6 repetitions, of 65% of 1RM	5–10 min	Increases
		Ballistic bench press throws: Back squats: 1 set, 1–3 repetitions, at 100% 3RM	8–12 min	Increases
Intercollegiate	Basketball, volleyball, handball, soccer	Back half squats: 1set, 10 repetitions, at 90% of 1RM	5 min	Increases 10-m, 20-m sprints
Intercollegiate	Football	Back squat: 1 set, 3 repetitions, at 90% of 1RM	4 min	Increases 40-m sprint
	Track sprinters	Back squat: 5 sets, 1–3 repetitions, at 90% of 1RM	5–10 min	Increases 20-m sprints
Elite	Sprint cyclists	Parallel back squats: 10 sets, 1 rep, at 90% of 1RM	5 min	Increases 10-s sprint
Elite	Triathletes (distance runners)	10-s maximum isometric elbow extensors and ankle plantar flexors	5 min	Increases
Elite	Soccer	Dead lift: 1 set, 5 repetitions, of 5RM	4–6 min	No significant increases, but large individual variation, PAP considered on individual basis
Elite	Soccer	Jump squats, Half squats countermovement jump: 3 sets, 5 repetitions, at 60% of 1RM	5–10 min	Increases
Women				
Collegiate	Soccer track sprinters	1 set, 4 repetitions, at 100% 4RM	9 min	Increases 100-m sprint
Elite	Jumpers	1 set maximum isometric squat, 3 trials on maximum 3-s Cybex knee squats at 120 and 130°	5 min	Increases
Elite	Hockey and softball	Half squats: 2 sets, 4 repetitions, at 30% of 3RM	5 min	Increases

women, use the same protocols as for men or they could possibly perform 1 set of 4 repetitions with a load of 4RM of 100% maximum with the parallel half back squat; and (e) optimal recovery period to maximize the PAP effect on PPO activities such as sprint and jump performances appears to be between 8 and 12 minutes for the lower body.

PRACTICAL APPLICATIONS

The majority of PAP warm-up research indicates that jump and sprint performances should increase after precise PAP warm-up protocols in elite male and female athletes. After a normal pre-competitive warm-up session, it is advised that strength coaches should precisely time the re-warm-up of a heavy preload with a parallel back half squat to elicit PAP before the competitive event as a means to increase their athletes' power. Based on the totality of PAP studies, the preload warm-up should include between 8 and 12 minutes of recovery before the sprint events to obtain the greatest PAP benefit. In addition, strength coaches should be aware that individual responses exist in terms of optimal recovery for PPO after a preload stimulus. Therefore, strength coaches should individually determine their athletes' optimal recovery for PPO to improve sprint times.

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